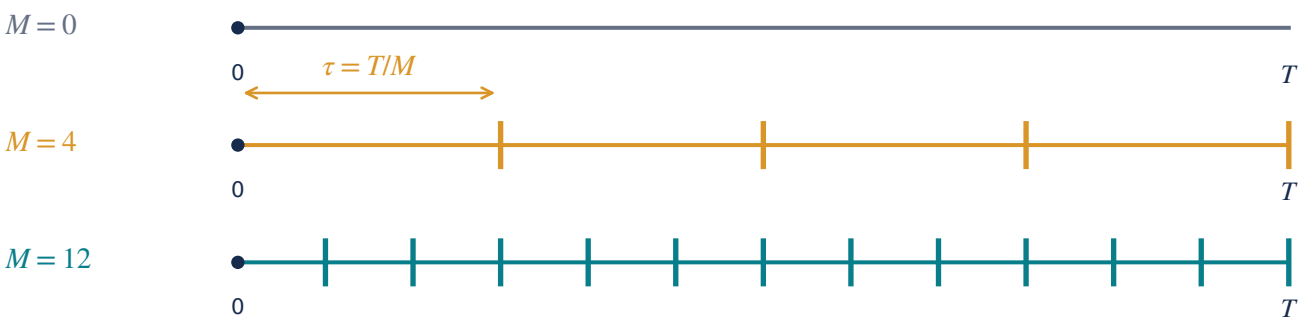
01 Measurement protocol

Ideal instantaneous binary projections; outcomes discarded unless explicitly post-selected.

$\rho_0 = |00\rangle\langle 00|$ → $U(\tau) = e^{-iH\tau/\hbar}$ → $\mathcal{D}(\rho) = P\rho P + Q\rho Q$



$P = |00\rangle\langle 00|, \quad Q = I - P$ $\qquad$ $p_P(T) = \mathrm{Tr}[P\rho(T)]$

Non-selective: sum all outcomes. All-success: $S_M = |\langle 00|U(T/M)|00\rangle|^{2M}$. Events include T.